

Daniel Grimmer

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Research Interests

Areas of Specialization: Philosophy of Physics, Philosophy of Science, Philosophy of Biology, Metaphysics of Space and Time, Evolutionary Epistemology, Epistemology of Artificial Intelligence, Quantum Field Theory, Quantum Information, General Relativity.

Areas of Competence: Ethics of Artificial Intelligence, Philosophical Logic, Philosophy of Mathematics, Early Modern Philosophy, Thermodynamics.

Academic Employment and Education

Postdoctoral Associate in Philosophy July 2025 – June 2027
Yale University, New Haven CT, USA

Hertz Postdoctoral Fellowship in History and Philosophy of Physics Oct 2024 – June 2025
University of Bonn, Bonn, Germany

DPhil in Philosophy Sept 2021 – Aug 2024
University of Oxford, Oxford, UK

Thesis: *Searching for New Spacetimes: Towards a Dynamics-First View of Topology.*

Passed viva with no corrections.

MSt in Philosophy of Physics Sept 2020 – June 2021
University of Oxford, Oxford, UK

Visiting Postdoctoral Fellow, Barrio RQI Sept 2020 – May 2022
University of Waterloo, Waterloo ON, Canada

PhD in Physics (Quantum Information) Sept 2015 – Sept 2020
University of Waterloo, Waterloo ON, Canada

Thesis: *Interpolated Collision Model Formalism.*

BA in Mathematics Sept 2011 – May 2015
University of Oklahoma, Norman OK, USA

BS in Engineering-Physics Sept 2011 – May 2015
University of Oklahoma, Norman OK, USA

Papers Currently Under Review (2)

1. **Title:** Innateness In Silico at Scale: How Evolutionary Meta-Learning Repositions Neural Networks within the Nativism–Empiricism Debate

Author: [Daniel Grimmer](#)

Journal: Synthese (Submitted Mar 2026; Revise and Resubmit) **Preprint:** PhilSci:28373

2. **Title:** Direct From Darwin: Deriving Advanced Optimizers From Evolutionary First Principles

Author: [Daniel Grimmer](#)

Peer-Reviewed Publications and DPhil/PhD Theses (18)

1. **Title:** Equivalence and Determinism in Light of Topologically-Induced Structure
Authors: [Daniel Grimmer](#), James Read
Journal: Erkenntnis (Accepted Jul 2026; in production) **Preprint:** PhilSci:29119
2. **Title:** Searching for New Spacetimes: Towards a Dynamics-First View of Topology
Author: [Daniel Grimmer](#)
D.Phil. Thesis, University of Oxford (Submitted Jul 2024, Viva Aug 2024)
Outcome: Passed Viva with No Corrections
3. **Title:** Dualities, Quantum Mechanics, and the Uncommon Common Core
Authors: [Daniel Grimmer](#), Enrico Cinti, Rasmus Jakslund
Journal: British Journal for the Philosophy of Science (Mar 2024)
4. **Title:** The Pragmatic QFT Measurement Problem and the Need for a Heisenberg-like Cut in QFT
Author: [Daniel Grimmer](#)
Journal: Synthese 202, 104 (Sept 2023) **Preprint:** arXiv:2205.09608
5. **Title:** Decoding Quantum Field Theory with Machine Learning
Authors: [Daniel Grimmer](#), Irene Melgarejo-Lermas, José Polo-Gómez, Eduardo Martín-Martínez
Journal: Journal of High Energy Physics 2023, 031 (Aug 2023) **Preprint:** arXiv:1910.03637
6. **Title:** Measurements in QFT: Weakly Coupled Local Particle Detectors and Entanglement Harvesting
Authors: [Daniel Grimmer](#), Bruno de S. L. Torres, Eduardo Martín-Martínez
Journal: Physical Review D 104, 085014 (Oct 2021) **Preprint:** arXiv:2108.02794
7. **Title:** The Unruh Effect in Slow Motion
Authors: Silas Vriend, [Daniel Grimmer](#), Eduardo Martín-Martínez
Journal: Symmetry 13(11), 1977 (Oct 2021) **Preprint:** arXiv:2011.08223
8. **Title:** Dimensional Reduction of Cavities with Axial Symmetry: Are Optical Fibers Really One-Dimensional?
Authors: [Daniel Grimmer](#), Richard Lopp, Eduardo Martín-Martínez
Journal: Physical Review A 104, 013723 (Jul 2021) **Preprint:** arXiv:2104.00745
9. **Title:** Interpolated Collision Models Formalism
Author: [Daniel Grimmer](#)
Ph.D. Thesis, University of Waterloo (Sept 2020) **Preprint:** arXiv:2009.10472
10. **Title:** Collisional Quantum Thermometry
Authors: Stella Seah, Stefan Nimmrichter, [Daniel Grimmer](#), Jader Santos, Valerio Scarani, Gabriel Landi
Journal: Physical Review Letters 123, 180602 (Oct 2019) **Preprint:** arXiv:1904.12551
11. **Title:** Zeno Friction and Antifriction from Quantum Collision Models
Authors: [Daniel Grimmer](#), Achim Kempf, Robert B. Mann, Eduardo Martín-Martínez
Journal: Physical Review A 100, 042702 (Oct 2019) **Preprint:** arXiv:1902.07738
12. **Title:** A Classification of Markovian Fermionic Gaussian Master Equations
Authors: Marvellous Onuma-Kalu, [Daniel Grimmer](#), Robert B. Mann, Eduardo Martín-

Martínez

Journal: Journal of Physics A 52, 435302 (Oct 2019)

Preprint: arXiv:1902.02239

13. **Title:** Thermal Contact: Mischief and Time Scales

Authors: Daniel Grimmer, Robert B. Mann, Eduardo Martín-Martínez

Journal: Journal of Physics A 52, 395305 (Sept 2019)

Preprint: arXiv:1805.11118

14. **Title:** Gaussian Ancillary Bombardment

Authors: Daniel Grimmer, Eric Brown, Achim Kempf, Robert B. Mann, Eduardo Martín-Martínez

Journal: Physical Review A 97, 052120 (May 2018)

Preprint: arXiv:1802.08629

15. **Title:** A Classification of Open Gaussian Dynamics

Authors: Daniel Grimmer, Eric Brown, Achim Kempf, Robert B. Mann, Eduardo Martín-Martínez

Journal: Journal of Physics A 51, 245301 (May 2018)

Preprint: arXiv:1709.07891

16. **Title:** Purification in Rapid Repeated Interaction Systems

Authors: Daniel Grimmer, Robert B. Mann, Eduardo Martín-Martínez

Journal: Physical Review A 95, 042114 (Apr 2017)

Preprint: arXiv:1611.07530

17. **Title:** Open Dynamics under Rapid Repeated Interaction

Authors: Daniel Grimmer, David Layden, Robert B. Mann, Eduardo Martín-Martínez

Journal: Physical Review A 94, 032126 (Sept 2016)

Preprint: arXiv:1605.04302

18. **Title:** Quantum Phases of Soft-Core Dipolar Bosons in Optical Lattices

Authors: Daniel Grimmer, Arghavan Safavi-Naini, Barbara Capogrosso-Sansone, Şebnem Güneş Soyler

Journal: Physical Review A 90, 043635 (Oct 2014)

Preprint: arXiv:1408.3532

In Preparation (1)

1. **Title:** Das Neue Raumproblem: The Phenomenology of Spacetime Dualities and Evolutionary Epistemology

Author: Daniel Grimmer

Unpublished Preprints (5)

1. **Title:** Spacetime Representation Theory: Setting the Scope of the ISE Method of Topological Redescription

Author: Daniel Grimmer

Status: Withdrawn from review; reworked into *Das Neue Raumproblem* (Jun 2023)
Preprint: arXiv:2306.08110

2. **Title:** From Humean Laws to a Neo-Kantian Spacetime: A Dynamics-First View of Topology

Author: Daniel Grimmer

Preprint only (Aug 2023)

Preprint: arXiv:2308.14146

3. **Title:** Introducing the ISE Methodology: A Powerful New Tool for Topological Redescription

Author: Daniel Grimmer

Preprint only (Mar 2023)

Preprint: arXiv:2303.04130

4. **Title:** A Discrete Analog of General Covariance, Part 2: Despite What You've Heard, a Perfectly Lorentzian Lattice Theory

Author: Daniel Grimmer

Preprint only (May 2022)

Preprint: arXiv:2205.07701

5. **Title:** A Discrete Analog of General Covariance, Part 1: Could the World Be Fundamentally Set on a Lattice?
Author: Daniel Grimmer
Preprint only (Apr 2022) **Preprint:** arXiv:2204.02276

Journal Refereeing

Australasian Journal of Philosophy · British Journal for the Philosophy of Science · Foundations of Physics · Annals of Physics · Physical Review Research · Physical Review A

Teaching Experience (13 Years)

Lecturer 2025 – 2027

Department of Philosophy, Yale University

Philosophy of Physics: Space, Time, and Quantum 2026–27

Self-designed, 20 students

First-Order Logic 2026–27

Large class, 121 students

AI Epistemology and Ethics: Knowing Machine Morals 2025–26

Self-designed, 20 students

First-Order Logic 2025–26

Large class, 121 students

Oxford terms: MT = Michaelmas (autumn), HT = Hilary (winter), TT = Trinity (spring).

Lecture Series MT 2023

Faculty of Philosophy, University of Oxford

Metaphysics of the Laws of Nature MT 2023

Self-designed, 10 students

Tutorial Teacher MT 2021 – TT 2023

Various colleges, University of Oxford

Tutored philosophy undergraduates one-on-one and in groups of up to seven; set and marked weekly 2,000-word essays and logic problem sheets; held additional office hours.

Philosophy of Time Pembroke College, TT 2023

OxNet Outreach

Leibniz–Clarke Correspondence Magdalen College TT 2023

Philosophy of Mathematics Brasenose College TT 2023

Elements of Deductive Logic St Hugh's College HT 2023

Philosophy of Spacetime Wycliffe Hall MT 2022

Philosophy of Science St Hilda's College HT 2022

Introduction to Logic Oxford Study HT 2022

Abroad Programme

Philosophy of Time Oriel College, HT 2022

Academic Taster

Elements of Deductive Logic Pembroke College HT 2022

Philosophy of Spacetime	Exeter College	HT 2022
Philosophy of Science	St Catherine's College	HT 2022
Introduction to Logic	Pembroke College	MT 2021

Teaching Assistant in Philosophy MT 2022 – HT 2023
Various colleges, University of Oxford

General Philosophy	Oriel College	HT 2023
General Philosophy	St Hugh's College	MT 2022 – HT 2023

Teaching Assistant in Physics Sept 2015 – Dec 2017
University of Waterloo

Led classes of up to 30 students; set and marked exams; prepared lecture notes.

Physics 1	2015–17
Mechanics	2015–17
Electrical Engineering 1	2015–17
Quantum Physics 1	2015–17
Quantum Theory 2	2015–17

Mathematics Tutor Sept 2013 – May 2015
University of Oklahoma

One-on-one tutoring for undergraduates in the university's open-door walk-in study area. Subjects: college algebra, pre-calculus, trigonometry, calculus and analytical geometry (I–IV), differential and integral calculus (I–III), ordinary differential equations, and linear algebra.

Children's Mathematics Tutor May 2014 – Jul 2014
Mathnasium OKC

One-on-one teaching with students aged 4–18, some with dyscalculia (i.e., mathematical dyslexia). Learned to teach students of widely varying background by meeting them at their level.

Academic Funding Secured

Clarendon Scholarship & Oxford-Reuben Graduate Scholarship	Oxford	£142,050
Pembroke College Alumni & Cooper Trust Scholarship	Oxford	£13,200
Vanier Scholarship	Waterloo	\$150,000 CAD
OU Regents' Scholarship	Oklahoma	\$50,600 USD
Astronaut Scholarship	Oklahoma	\$10,000 USD

Talks and Conference Presentations (since 2021)

<i>How Evolutionary Meta-Learning Repositions Neural Networks in the Nativism-Empiricism Debate</i> British Society for the Philosophy of Science (BSPS)	Leeds, July 2026
<i>Topological Asymmetry Neutralizes Leibniz's Shift Argument and Collapses Categorical Equivalence</i> Foundations of Physics, University of California, Irvine	Irvine CA, June 2026
<i>Hausdorff and the Underdetermination of Space by Evidence</i> Felix Hausdorff Conference, Yale University	New Haven, May 2026
<i>Evolutionary Epistemology In Silico: Exploring Nature and Nurture with Neural Networks</i> Laurie Paul's Group Meeting, Yale University	New Haven, Apr 2026
<i>Evolutionary Epistemology In Silico: Exploring Nature and Nurture with Neural Networks</i> ELLMM Talk Series, Yale University	New Haven, Apr 2026
<i>Evolutionary Meta-Learning: A New Framework for Linguistic Nativism</i> Computational Linguistics at Yale (CLAY)	New Haven, Feb 2026
<i>The End of Nature vs. Nurture: Why AI Won't Inherit Our Philosophical Puzzles</i> P&P Meeting, Digital Ethics Center, Yale University	New Haven, Sept 2025
<i>Undercutting the Spatiotemporal Underpinnings of the Humean Mosaic</i> Quantum Gravity and the Laws of Nature Workshop	Geneva, Dec 2024
<i>Searching for New Spacetimes: The ISE Method of Topological Redescription</i> History and Philosophy of Physics Research Seminar, University of Bonn	Bonn, Nov 2024
<i>What Are the Laws of Nature? And What Do They Do?</i> Surrey Summer School	Southampton, July 2024
<i>In Search of New Spacetimes: The ISE Method of Topological Redescription</i> Philosophy of Physics Seminar, University of Oxford	Oxford, May 2024
<i>In Search of New Spacetimes: A Dynamics-First View of Topology</i> Job talks — Technion, University of Bonn, and Chapman University	Feb 2024
<i>In Search of New Spacetimes: A Dynamics-First View of Topology</i> Philosophy of Physics Workshop, University of Oxford	Oxford, Sept 2023
<i>Deflating Spacetime: A Dynamics-First View of Topology</i> Third Hermann Minkowski Meeting on the Foundations of Spacetime Physics	Albena, Sept 2023

<i>From Humean Laws to a Kantian Spacetime: A Dynamics-First View of Topology</i> First International Colloquium: ‘Emergence and Time’	Braga, Sept 2023
<i>A Dynamics-First View of Topology: Topological Redescription via the ISE Methodology</i> Foundations of Physics Conference 2023	Bristol, July 2023
<i>Regularity Relationism without a Manifold</i> Philosophy of Logic, Mathematics and Physics Graduate Conference	London ON, June 2022
<i>The Pragmatic QFT Measurement Problem and the Need for a Heisenberg-like Cut in QFT</i> Quantum Field Theory on Curved Spacetime 2022	Online, May 2022
<i>The Pragmatic QFT Measurement Problem Requires a Heisenberg-like Cut in QFT</i> Establishing the Philosophy of Supersymmetry 2022	Stockholm, May 2022
<i>A Discrete Analog of General Covariance: Could the World Be Fundamentally Set on a Lattice?</i> Irvine–London–Munich–PoliMi–Salzburg Conference in Philosophy and Foundations of Physics	Milan, April 2022
<i>The Unruh Effect in Slow Motion</i> RQI-Online Conference 2021	Waterloo, March 2021

Service and Governance

- **Seminar Organizer**, Graduate Student Lecture Series (“Pop-grunch”), March 2022 – January 2023.
- **Graduate Student Representative**, Humanities Division Board, University of Oxford, Sept 2020 – Oct 2021. Chair of the Graduate Joint Consultative Committee; also served on the Humanities Divisional Board, Graduate Studies Committee, Education Development Panel, and Committee for Library Provisions.
- **Director**, Board of Directors, Graduate Student Association, University of Waterloo, May 2019 – Aug 2020. A not-for-profit corporation with approximately 20 full-time employees and a \$1M CAD annual operating budget. Chief Returning Officer, March – April 2019.
- **President**, Graduate Student Association, Institute for Quantum Computing, April 2018 – Oct 2019; Physics Graduate Student Association, Nov 2018 – Nov 2019.
- **Lead Conference Organizer**, Canadian Graduate Quantum Conference, 2019 and 2020.